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Roll No

ME-7003-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

OR and Supply Chain

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) List essential characteristics of a Linear programming model.
b) What are the advantages of Linear programming method?
2. Maximize $Z = 3x_1 + 2x_2 + 5x_3$
Subject to $x_1 + x_2 + x_3 \leq 9$
 $2x_1 + 3x_2 + 5x_3 \leq 30$
 $2x_1 - x_2 - x_3 \leq 8$
 $x_1, x_2, x_3 \geq 0$
3. a) Discuss importance of Supply chain.
b) What is Forecasting? State its importance.
4. a) What is EOQ? Discuss its significance.
b) Explain ABC and VED analysis of inventory.

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5. A drive in bank window has a mean service time of 2 min while the customers arrive at a rate of 20 per hour. Assuming that there represent rates with a Poisson distribution. Find:
 - a) What percentage of time will the teller be idle
 - b) After driving up how long will it take the average customer to wait in line and be served and
 - c) What fraction of customer will have to wait
6.
 - a) What do you understand by Competition strategy? Explain.
 - b) What is a Zero sum game?
7.
 - a) What are the benefits of decision under certainty?
 - b) Define heuristic and metaheuristic algorithms.
8. Write short notes on any two
 - a) Transportation problem
 - b) E-business model
 - c) Lean manufacturing
 - d) Travelling salesman problem
